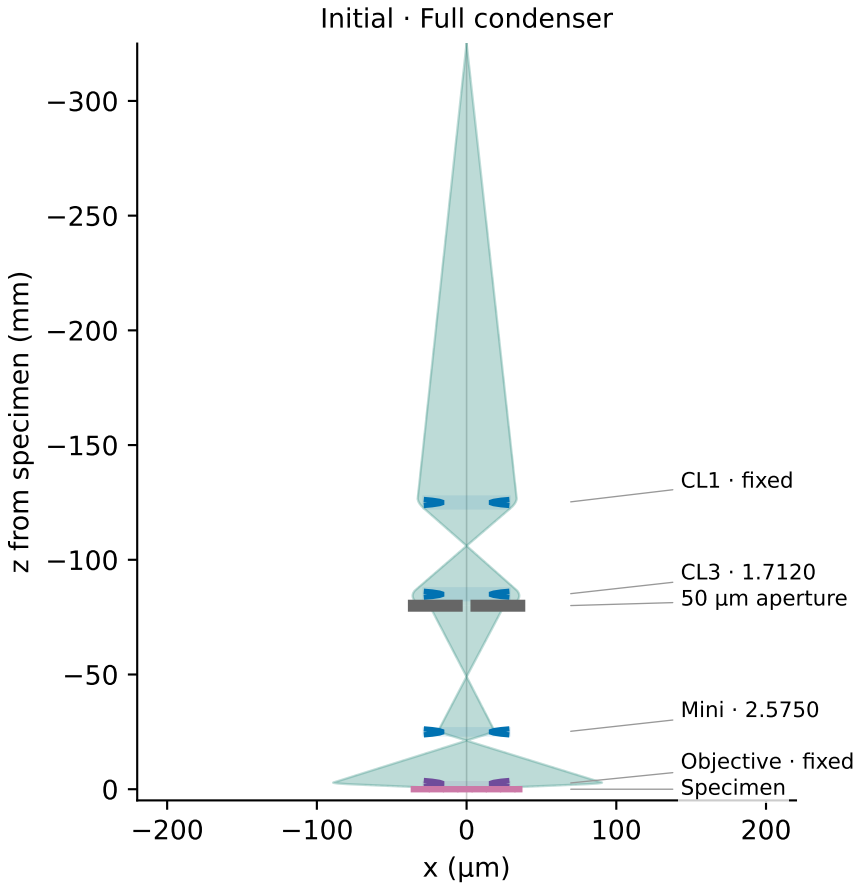
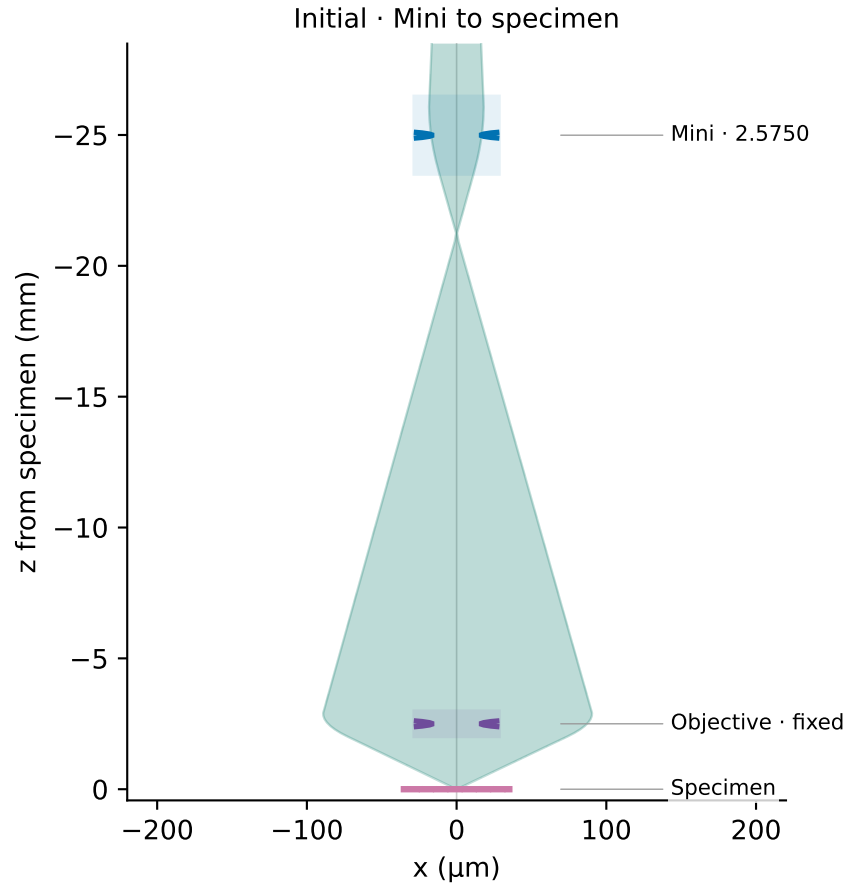
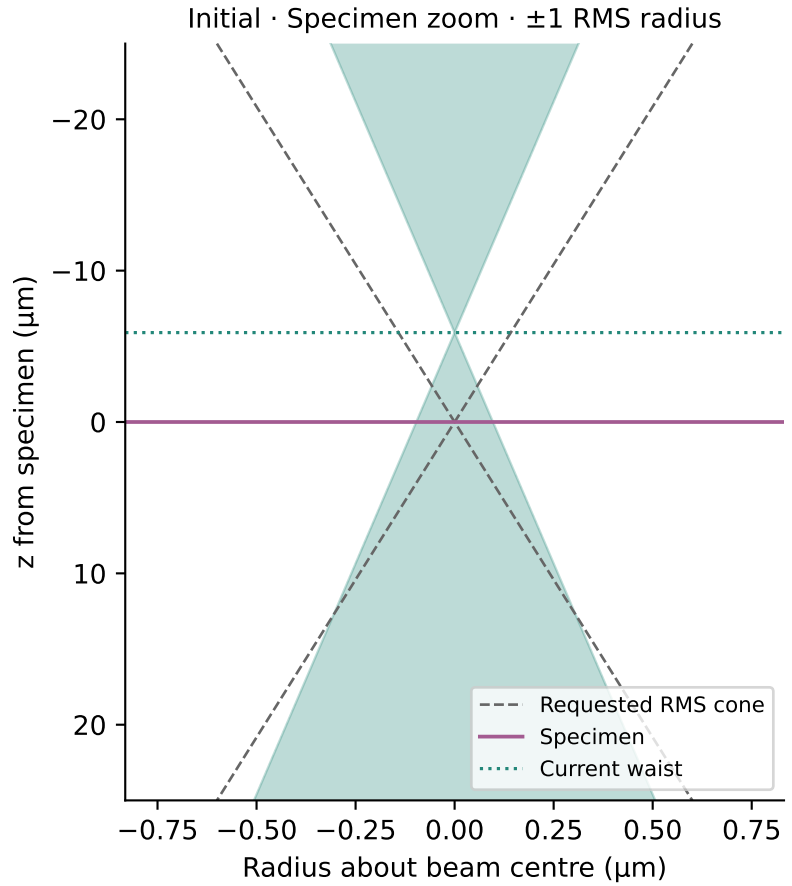
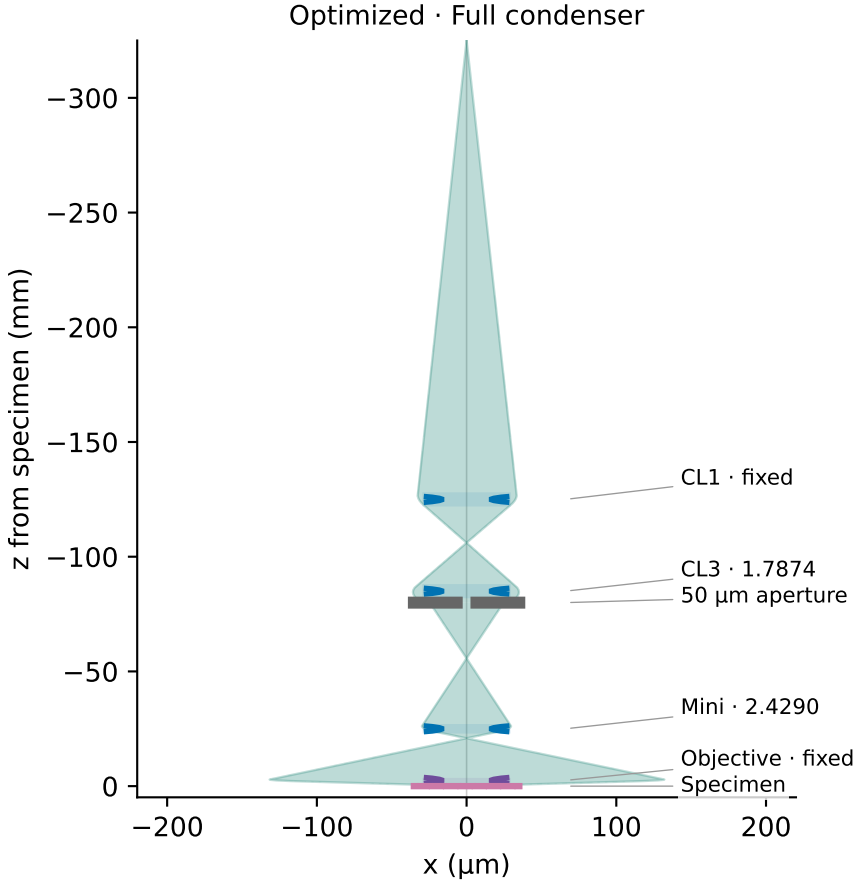
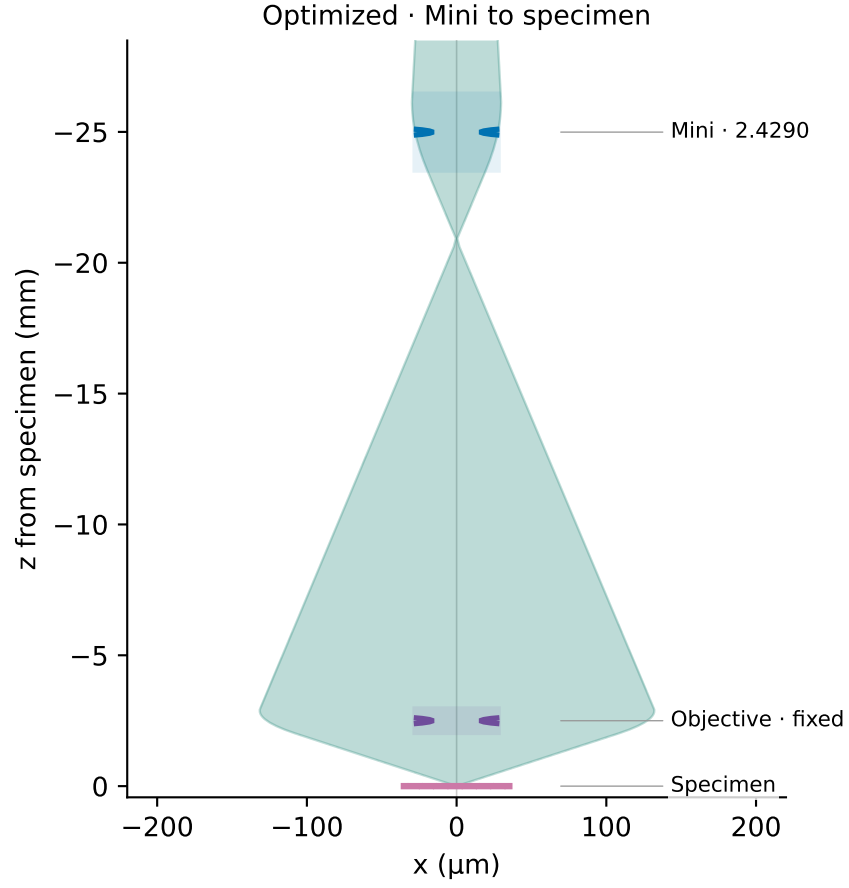
Gradient descent reshapes the beam to 24 mrad RMS at the specimen

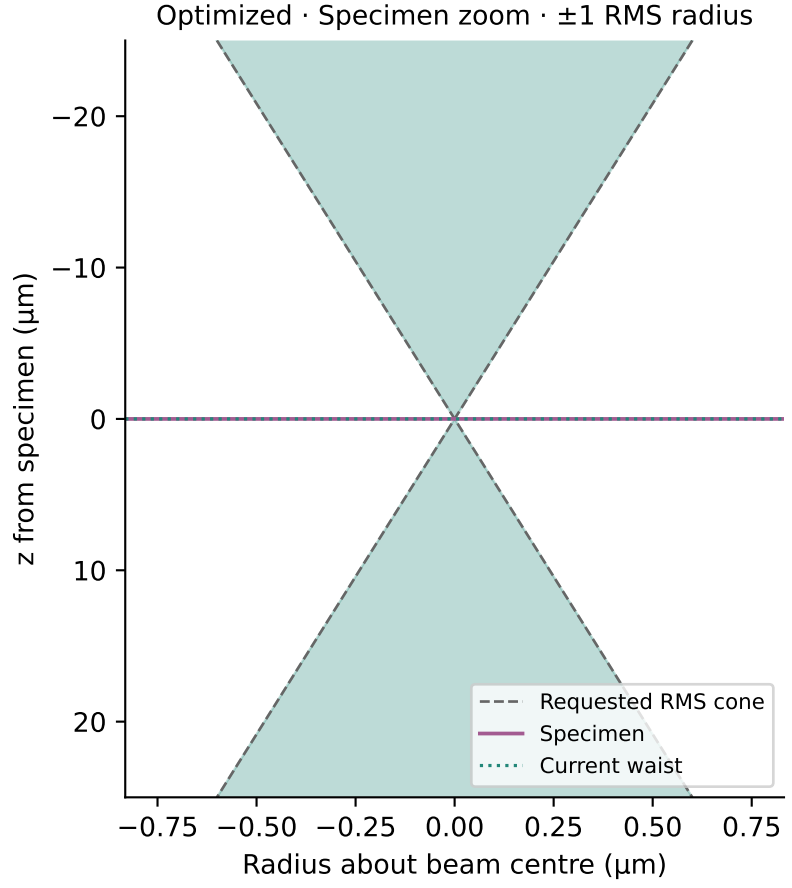












Start: 16.388 mrad, waist -5.906 µm → Final: 24 mrad, waist at specimen
Left: native clipped ray bundle · right: smooth-aperture RMS envelope · fixed axes for comparison